ORGANIC CUMULATIVE EXAMINATION

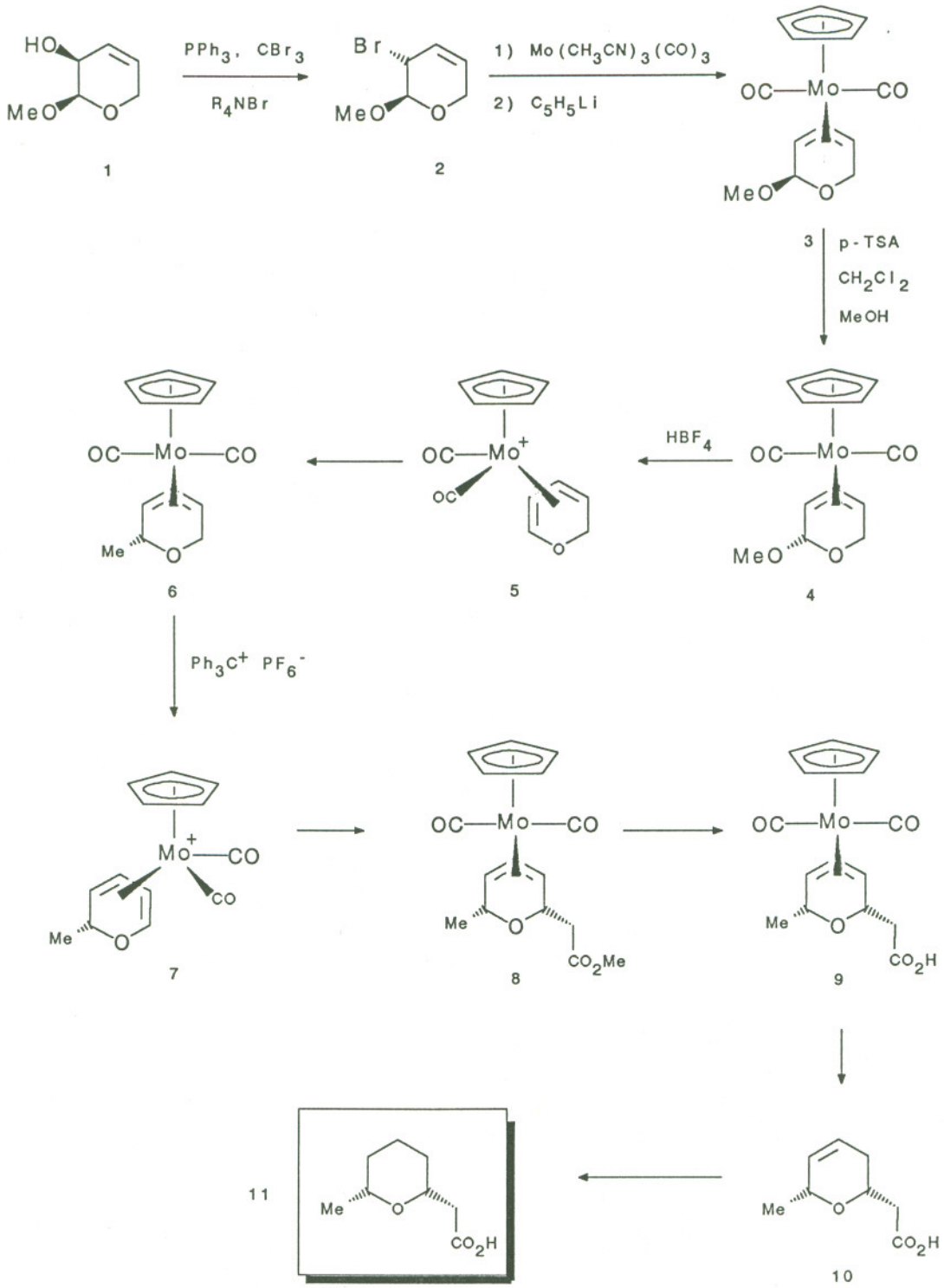
January 19, 1991

The reaction scheme shown on the next page was recently reported in the Journal of the American Chemical Society (Hansson, S.; Miller, J. F.; Liebeskind, L. S. *J. Am. Chem. Soc.* **1990**, *112, 9660*). The product **11** is the enantiomer of a component of the scent gland secretion of *Viverra Civetta*.

Answer *four* of the following questions in as much detail as possible.

1. Give a mechanism of reaction for the first step in the sequence (**1** to **2**)
2. Give a mechanism for the transformantion of **4** to **5**. It is interesting to note that reaction of **3** under the same conditions gives poor yields of **5**. Why?
3. Suggest a reagent which would accomplish the **5** to **6** reaction.
4. Give a mechanism and explain the selectivity of the **6** to **7** reaction.
5. Suggest a reagent to accomplish the **7** to **8** reaction.
6. Give a mechanism for the **8** to **9** transformation.

Continued...


HO
MeO
O
1
PPh3, CBr3
R4NBr
Br
MeO
O
2
1) Mo(CH3CN)3(CO)3
2) C5H5Li
OC—Mo—CO
MeO
O
3
p-TSA
CH2Cl2
MeOH
OC—Mo—CO
MeO
O
4
HBF4
OC—Mo+
OC
O
5
OC—Mo—CO
Me
O
6
Ph3C+ PF6-
Mo+—CO
CO
Me
O
7
OC—Mo—CO
Me
O
CO2Me
8
OC—Mo—CO
Me
O
CO2H
9
Me
O
CO2H
10
11
Me
O
CO2H